



FMCG

Fast Moving Consumer Goods

CSCI 112: Contemporary Databases
Final Project

Group 3

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Fast-Moving Consumer Goods

In the fast-moving consumer goods (FMCG) industry, **a point of sales system streamlines the transaction process** when customers were to purchase a product. It is also **automates the process of updating inventory**, whenever a product is purchased.



Functionalities

Processing Retail Transactions

The POS system is utilised when customers purchase an item from a physical supermarket/grocery store.



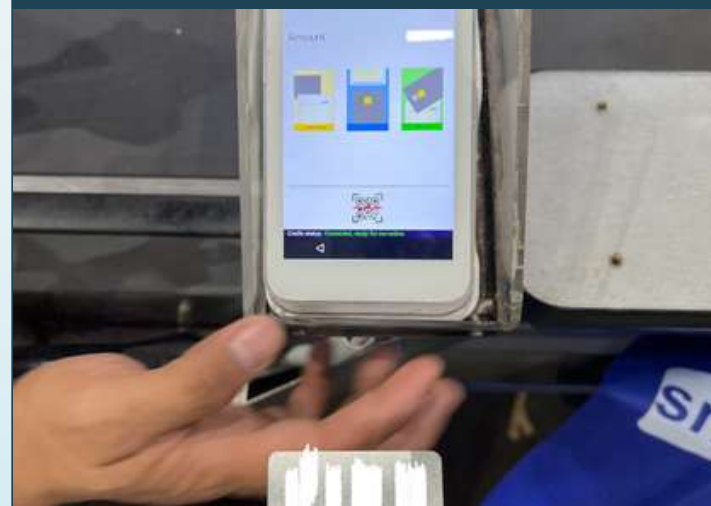
Managing Inventory Supply

The system retrieves and updates product information such as stock and price levels to provide accurate and up-to-date data.



Handling Payment

Various payment methods are processed and involve secure connections to payment gateways. Cash is also accepted.



Reporting Sales

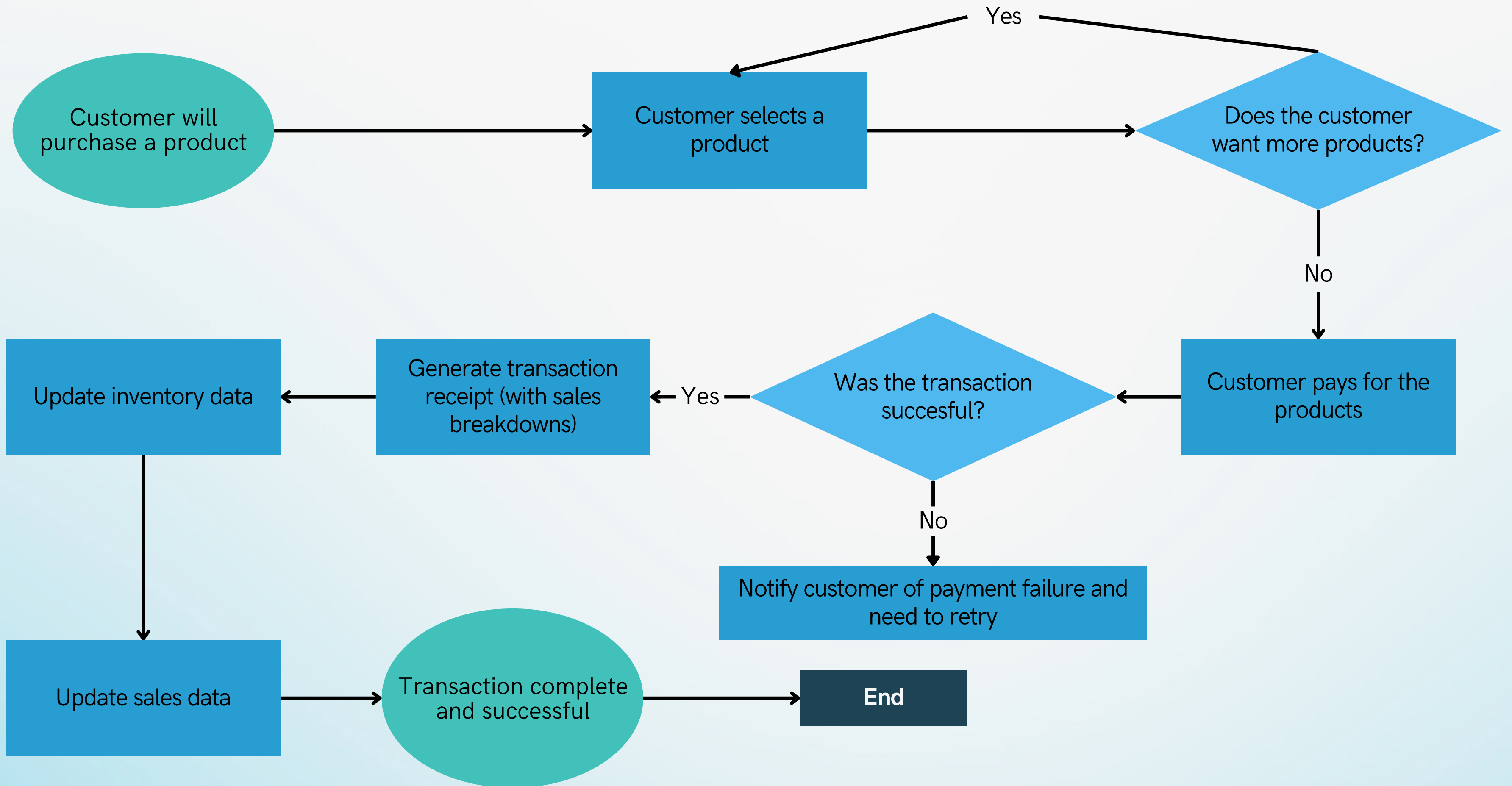
Sales transaction data help FMCG companies monitor sales performance KPIs and inventory requirements.



Enhancing Customer Experience

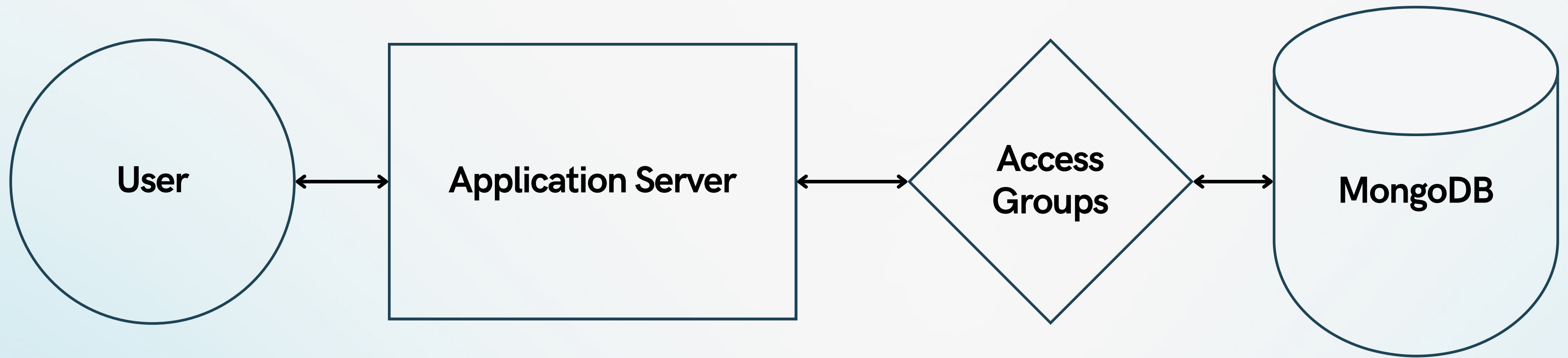
The seamless and efficient transaction process improves customer satisfaction in groceries.



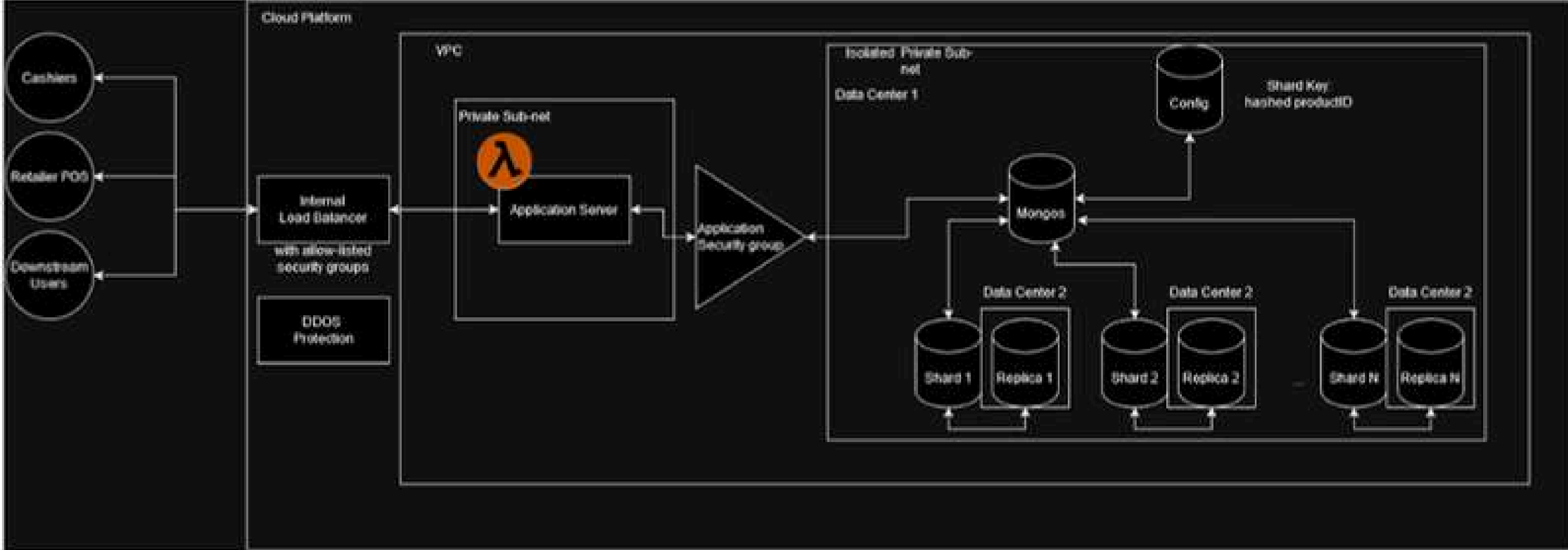


General Business Process





**Actual
Architecture**



Ideal Architecture

SCHEMA DESIGN

01

Product

- "_id": ObjectID
- "productId": int
- "productName": string
- "productType": string
- "productDescription": string
- "productPrice": int
- "supplier": string
- "productStock": int
- "lastUpdate": ISODate

02

Transaction

- "_id": ObjectID
- "items": itemsArray
 - "productId": int
 - "productName": string
 - "quantity": int
 - "price": int
- "totalPrice": int
- "transactionDate": ISODate
- "cashierName": string

Access Patterns (CRUD)

Create

- **Create Product**
 - The user creates a new “product” document and writes it into the database.
 - Used when adding new items to be sold in the supermarket.
- **Create Transaction**
 - The user creates an “invoice” document and writes it in the database.
 - Used during a customer is making a purchase; to report a summary of the products ordered and its total price.

Update

- **Update Product**
 - The user updates an existing “product” document in the database with new details.
 - Use cases include:
 - Product offering has been changed and needs to be updated in the database.
 - A customer purchased a product, which updates the available stock of the said product.
- **Update Transaction**
 - The user updates an existing “invoice” document in the database with new details.
 - Used when an invoice needs to be updated potentially due to an error in its creation.

Delete

- **Delete Product**
 - The user deletes a “product” document from the database.
 - Used when a product is no longer being offered.
- **Delete Transaction**
 - The user deletes a “invoice” document from the database.
 - Used when an invoice is no longer necessary.

Aggregation Pipelines

Market Leader

Obtain the highest selling products from the list of items being sold in a grocery.

Top Suppliers

Obtain a list of the top 5 suppliers who provide the most quantity of items.

Inventory Restocking

Identify products that need to be restocked based on the remaining inventory stock.

Sales Growth

Calculate the percentage change of the quantity sold of an item between two time periods.



Demonstration

1. CRUD
2. Aggregation Pipelines



Thank you!